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Unlocking the Magic of *Pithecellobium dulce*

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Abstract

The bark and pulp of Manila Tamarind is used as a traditional remedy against gum ailments, toothache and hemorrhage. Bark extract is also used against dysentery, diarrhea and constipation. An extract of leaves is used for gall bladder ailments and to prevent miscarriage. Seeds when grounds are used to cleanse ulcers. Numerous studies have been performed on anti-oxidant, anti- inflammatory, anti- diabetic, anti-cancer properties of Manila tamarind. It provides relief from pain, eczema, fever, cold, sore throat, pigmentation, acne and pimples.

Keywords: *Pithecellobium dulce,* Medicinal and Health benefits.

Introduction

It originated from Mexico, then went to America, Central Asia and then to India. Although, these trees have been seen all along the highways in India, no one knew about its culinary use. It resembles tamarind and is widely called as Manila Tamarind.

Morphological description:

Botanically it's called *Pithecellobium dulce* and it belong to the family Fabaceae

Common names: Vilayati imli, Jungli jilebi, Ingraji chinch, Manila Tamarind, Monkey pod, Madras thorn.

The bark of *P. dulce* is grey in color, becomes rougher, and begins to peel when mature. The size of the leaves is 2-2.5 × 1-2 cm, with kidney-shaped lobules and a pair of two leaves. Each leaf has a 2-15 mm slender spine at the root. Hairy crown blooms and little whiteheads with a diameter of 1 cm can be seen in P. dulce. The calyx of a joint tube surrounds 50 sparse stamens in the flower. When ripe, each pod is 10-15 × 1.0 cm long, which is spiral in shape and reddish brown in colour.
Botanically it's called *Pithecellobium dulce* and it belong to the family Fabaceae.

Table 1. Nutritional Value of Pithecellobium dulce

Component	Amount
Energy	78 kcal
Water	77.8 %
Protein	3 %
Fat	4 %
Carbohydrate	18.2 %
Fiber	1.2 %
Ash	6 %
Calcium (1.3 % RDI)	13 mg
Phosphorous (4.2 % RDI)	42 mg
Iron (2.7 % RDI)	5 mg
Sodium	19 mg
Potassium (6.3 % RDI)	222 mg
Vitamin A	15 mg
Thiamin/B1 (16.6 % RDI)	24 mg
Riboflavin/B2 (5.8 % RDI)	10 mg
Niacin/B6 (3 % RDI)	60 mg
Vitamin C (221 % RDI)	133 mg

Fig 1: ***Pithecellobium dulce*** **tree**

Fig 2. ***Pithecellobium dulce*** **leaves**

Fig 3: ***Pithecellobium dulce*** **fruits, flower and seeds**

Fig 4: Pithecellobium dulce bark

Manila tamarind contains:

Vitamin E - This contributes to aging.
Vitamin B1 - This helps to nourish the nerves and the brain.
Vitamin B2 - This contributes to the skin, nails and hair health.
Vitamin B3 (Niacin) - Which contributes to decrease levels of cholesterol.
Calcium - This helps to give a boost to bones and enamel.
Phosphorus - This contributes to the expansion and restoration of body.
Iron - This contributes to the prevention of fatigue of the body.

Health benefits

- Works as Antiseptic
- Lightens Skin
- Prevents Hair Loss
- Aids Weight Loss
- Regulates Blood Circulation
- Cures Acne and Pimples
- Removes Dark Spot
- Leaves - Remedy for indigestion
- Bark - curative for bowel movement/constipation
- Cures Acne and Pimples
- Removes Dark Spot

Medicinal uses

- **Anti-Inflammatory / Antibacterial:** Study of the fresh flowers of Pithecellobium dulce yielded a glycoside quercitin. The activity of the flavanol glycoside confirmed its anti-inflammatory antibacterial properties.

- **Antioxidant:** Study of the aqueous extract of Pithecellobium dulce leaves revealed phenolics including flavonoids and showed potent free radical scavenging activity

- **Anti-inflammatory:** Anti-inflammatory triterpene saponins of Pithecellobium dulce: A new bisdesmodictriterpenoid saponin, dulcin, was isolated from the seeds of PD.

- **Anti-Diabetic:** Study of ethanolic and aqueous leaf extract of P dulce in STZ-induced diabetic model in rats showed significant activity, aqueous more than the alcoholic extract, comparable to glibenclamide.

- **Antioxidant/ Antibacterial:** Study of fruit peel for antioxidant and antibacterial potential revealed significant activity in the ethyl acetate, methanolic and aqueous extracts.

- **Anti-Diabetic / Fruits:** Study evaluated the antidiabetic potential of P. dulce fruits in STZ-induced experimental diabetes in rats. Results showed significant reduction in blood glucose, glycosylated hemoglobin, urea and creatinine. There was also improved glycogen contact upon treatment with the extract.

- **Cardioprotective / Fruit Peel:** Study evaluated the effect of P. dulce peel in isoproterenol (ISO) induced myocardial infarction in adult male Wistar rats. ISO induced MI in rats showed increase in marker enzymes. Pretreatment PD fruit peel extracts positively altered the activities of marker enzymes and biochemical parameters in ISO-induced rats

- **Anthelmintic / Leaves:** Study evaluated leaf extracts of P. dulce in three different concentrations for anthelmintic activity against Pheretimaposthuma. The aqueous extract was more potent than the alcoholic extract, with activity comparable to the reference drug piperazine citrate.

Conclusion

Pithecellobium dulce (Roxb.) Benth, is a resourceful medicinal plant with a wide range of medicinal virtues that has availed widespread consideration recently. Almost all parts of the plant have immense nutritional value.

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